UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY (EXTERNAL)

Academic Year 2022 – 2nd Year Examination – Semester 3

IT 3206 - Data Structures and Algorithms

Part 2 - Structured Question Paper

(ONE HOUR)

To be completed by the candidate

BIT Examination Index No: .....

Important Instructions:

- The duration of the paper is **one (01) hour**.
- The medium of instruction and questions is English.
- This paper has **two** (**02) questions** and **08 pages**.
- **Answer all questions**
- **Write your answers** in English using the space provided **in this question paper**.
- Do not tear off any part of this answer book.
- Under no circumstances may a candidate remove this book, used or unused from the Examination Hall.
- Note that questions appear on both sides of the paper.
  If a page is not printed, please inform the supervisor immediately.
- All kinds of electronic devices, including calculators, are **not** allowed.
- *All Rights Reserved.*

Questions Answered

Indicate by a cross (×), (e.g. [X]) the numbers of the questions answered.

	Question numbers		
To be completed by the candidate by marking a cross (×).	1	2	
To be completed by the examiners:			

1) (a) One wants to reverse a String. Propose (i) a suitable data structure and (ii) a stepwise methodology for the reversing.

(2 Marks)

ANSWER IN THIS BOX

(i) Stack data structure

(ii) Push one by one character onto the stack

Pop from the stack one by one characters

(b) Write a methodology to simulate a linked list using two stacks.

Hint: You may use one stack as the list, and another stack can be used as temporary storage.

(2 Marks)

ANSWER IN THIS BOX

- To **add** an item at the head, simply push the item onto the stack.
- To **remove** from the head, pop from the stack.
- To **insert** into the middle somewhere, pop items from the "list" stack and push them onto the temporary stack until you get to your insertion point. Push the new item onto the "list" stack, then pop from the temporary stack and push back onto the "list" stack. Deletion of an arbitrary node is similar.

(c) If one evaluates the following pseudocode algorithm segment using a dry run, what would be the final content of the queue?

```
q = queue();
for(j=1; j<6;j++)
{
   q.enqueue (j);
}
for(i=1;i<6;i++)
{
   if(i%2 <>  0)
   {
   q.dequeue()
   }
   else{
   q.enqueue(i*3-1)
   }
}
```

(3 Marks)

ANSWER IN THIS BOX

contents of the queue =4,5,5,11

(d) List the difference between an Array and a Linked List on the following facts:

(i) Basic operations like insert, delete etc.
(ii) Space usage
(iii) Flexibility
(iv) Time usage
(v) Memory allocation

Note: One difference for each fact is sufficient for the answer.

(5 Marks)

ANSWER IN THIS BOX

Facts	linear Array	linked list
Basic operations like insert, delete etc.	Deletion and Insertions are difficult	Deletion and Insertions can be done easily.
Space usage	space is wasted	space is not wasted
Flexibility	It is difficult to reduce or extend according to requirements	It can be reduced or extended according to requirements
Time usage	It is expensive	It is not expensive
Memory Allocation	Elements are stored in consecutive memory locations	Elements may or may not be stored in consecutive memory locations

(e) One wants to create a max-heap using the following set of integers.

[8,23,9,13,45,21]

Nodes are inserted one by one, and any heap property violation is fixed when detected.

Which of the following are valid/invalid statements?

(i) During the max heap creation, 45 is swapped with 13
(ii) The total number of swaps during the heap creation is 5.
(iii) When creating the max heap, 45 is swapped with 13, and 13 is swapped with 08.
(iv) At the intermediate stage of heap construction, 13 is a root value.
(v) Nine (09) never becomes a root value during the heap construction

(5 Marks)

ANSWER IN THIS BOX

(i) True

(ii) True

(iii) False

(iv) False

(v) True

(f) The output of the breadth first traversal is given below:

6, 3, 15, 1,4,9,11

Consider the following algorithm

Step 1*: First, pick the first element of the array and make it root.*
Step 2*: Pick the second element, if its value is smaller than the root node value makes it left child,*
Step 3*: Else make it right child*

The recursive call in Step 2 and Step 3 continue to make the tree until all the nodes are picked. What would be the resultant tree if one applies the above data set to the algorithm?

(4 Marks)

ANSWER IN THIS BOX

Binary Search Tree

(g) What is the difference between recursion and iteration based on the following facts:

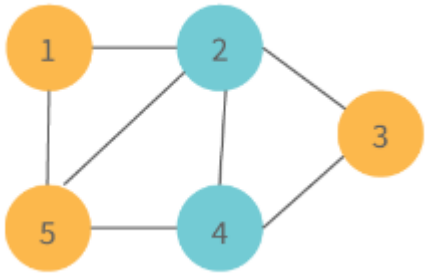
(i) Definition
(ii) Implementation
(iii) Memory Usage
(iv) Speed

(4 Marks)

ANSWER IN THIS BOX

Fact	Recursion	Iteration	
Definition	When a function calls itself directly or indirectly	When there are some set of instructions that are carried out repeatedly	
Implementation	Recursion is implemented with the use of function calls	Iteration is implemented by the use of loops	
Memory Usage	Stack memory is used to store local variables & parameters	No memory is used except for the initialization of control variables	
Speed	low speed of execution	High speed of execution	

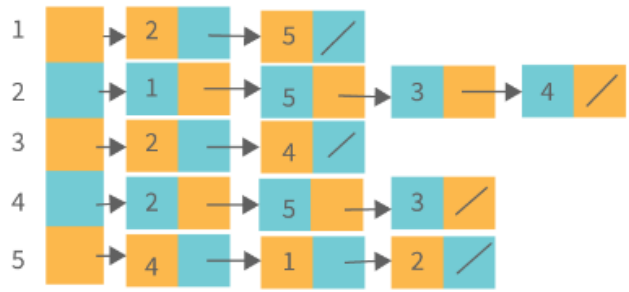
2 (a) Consider the following undirected graph.



(i) Draw the adjacency list of the above graph?

(3 Marks)

ANSWER IN THIS BOX



Or equivalent representation

(ii) Propose suitable data structures used to find the shortest path when traversing (i) Breadth First and (ii) Depth First in graphs.

(4 Marks)

ANSWER IN THIS BOX

(i) Breadth First traversal finds the shortest path using the queue data structure

(ii) **Depth First traversal finds the shortest path using the stack data structure**

(b) Suppose one sorts an array of ten integers using quicksort, and if he/she has just finished the first partitioning as follows:

15,24,52,79,61,85,72,95,62,105

List all possible pivot values that can be chosen to obtain the above intermediate result.

(4 Marks)

ANSWER IN THIS BOX

Answer 15,24,52

(c) The Merge Sort Algorithm consists of the following steps, which are unordered.

Step 1: *This division continues until the size of each sub array becomes 1*
Step 2: *The sub arrays are divided recursively.*
Step 3: *Unordered steps of the merge sort algorithm are given below:*
Step 4: *The merge procedure combines these trivially sorted arrays to produce a final sorted array.*
Step 5: *It divides the given unsorted array into two halves- left and right sub arrays.*
Step 6: *After each sub array contains only a single element, each sub array is sorted trivially.*
Step 7: *Then, the above discussed merge procedure is called.*

Write the correct order of the merge sort algorithm
Hint: Writing the step numbers is sufficient for the answer.

(5 Marks)

ANSWER IN THIS BOX

(a) Step 3, Step 5, Step 2, Step 1, Step 6, Step 7, Step 4

(d) Consider the following code segment.

```
int k=0;
for (int i=0;i<n;++) {
            if even(i) {for (int j=0;j<n;j++)
                                        k++;};
             else k*=2;
};
```

Prove the complexity of the algorithm as $O(n^2)$

(5 Marks)

ANSWER IN THIS BOX

$RT(n) = c + RT(outer_loop) * RT(body) = c + n * RT(body)$

$RT_{worst\text{-}case}(body) = c + \max\{ RT(inner_loop), c \} = c + n*c$

$RT(n) \leq n*RT_{worst\text{-}case}(n) = c+n*(c+n*c) \Rightarrow RT(n) = O(n^2)$

(e) Consider following time complexity of algorithms for any input of size n:

$T(n) = 2n^2+6n+5$

"When estimating running time of a relation for larger n values, there is no significant different when considering all the terms in the relation or considering the most dominant component of the relation"

Prove the validity of the above expression by substituting different values for n in the above relationship.

(4 Marks)

ANSWER IN THIS BOX

Consider T(n)=2n2+6n+5
n-0, $2n^2$=0, 6n=0, 5 then T(n) =5

n=10, $2n^2$=200, 6n=60, 5 then T(n)=265

n=100, $2n^2$=20000, 6n=600, 5 then T(n)=20665

n=1000 $2n^2$=2000000, 6n=6000, 5 then T(n)=2006005

For larger values of n

n=100000, $2n^2$=20000000000, 600000, 5 then T(n)=20,000,600,005

if we ignore all terms except the most dominant term, the difference in percentage is 0,0030%, so all other terms can be ignored for n's higher values. Therefore, the above expression is true.

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